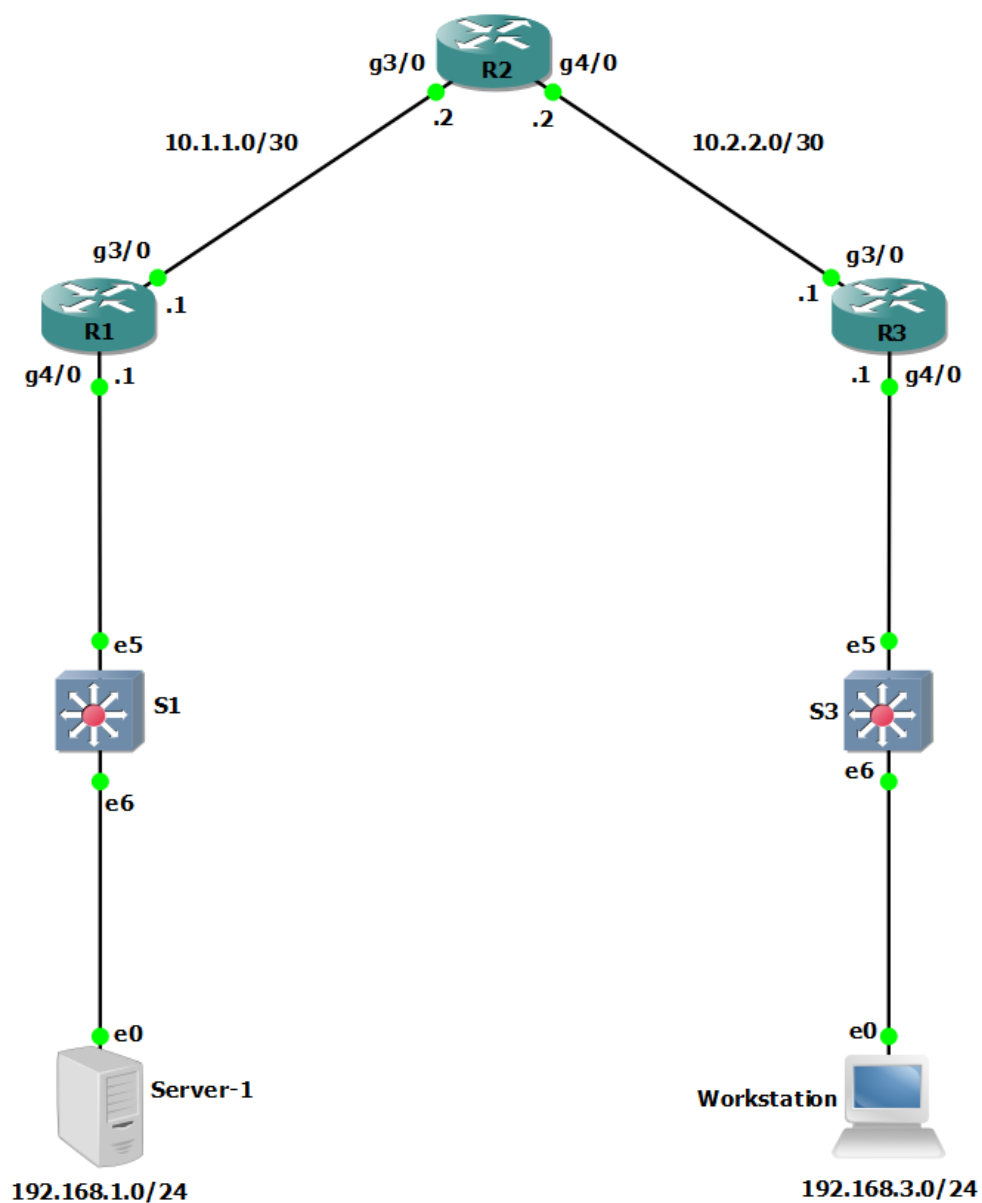


Lab - Configure Cisco IOS Resilience Management and Reporting (DAYUTH THY)

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G3/0	10.1.1.1	255.255.255.252	N/A	N/A
	G4/0	192.168.1.1	255.255.255.0	N/A	S1 E5
R2	G3/0	10.1.1.2	255.255.255.252	N/A	N/A
	G4/0	10.2.2.2	255.255.255.252	N/A	N/A
R3	G3/0	10.2.2.1	255.255.255.252	N/A	N/A
	G4/0	192.168.3.1	255.255.255.0	N/A	S3 E5
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 E6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 E6

Objectives

Part 1: Configure Basic Device Settings

- Cable the network as shown in the topology.
- Configure basic IP addressing for routers and PCs.
- Configure OSPF routing.
- Configure PC hosts.
- Verify connectivity between hosts and routers.

Part 2: Secure the Control Plane

- Configure OSPF Authentication using SHA256
- Verify OSPF Authentication

Background / Scenario

The router is a critical component in any network. It controls the movement of data into and out of the network and between devices within the network. It is particularly important to protect network routers because the failure of a routing device could make sections of the network, or the entire network, inaccessible. Controlling access to routers and enabling reporting on routers is critical to network security and should be part of a comprehensive security policy.

In this lab, you will build a multi-router network and configure the routers and hosts. You will configure administrative roles with different privilege levels.

Note: The routers used with hands-on labs are Cisco 4221 with Cisco IOS XE Release 16.9.6 (universalk9 image). The switches used in the labs are Cisco Catalyst 2960+ with Cisco IOS Release 15.2(7) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Before you begin, ensure that the routers and the switches have been erased and have no startup configurations.

Required Resources

- 3 Routers (Cisco 4221 with Cisco XE Release 16.9.6 universal image or comparable with a Security Technology Package license)
- 2 Switches (Cisco 2960+ with Cisco IOS Release 15.2(7) lanbasek9 image or comparable)
- 2 PCs (Windows OS with a terminal emulation program, such as PuTTY or Tera Term installed)
- Console cables to configure Cisco networking devices
- Ethernet cables as shown in the topology

Instructions

Part 1: Configure Basic Device Settings

In this part, set up the network topology and configure basic settings, such as interface IP addresses.

Step 1: Cable the network.

Attach the devices, as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each router.

- a. Console into the router and enable privileged EXEC mode.

```
Router> enable
```

```
Router# configure terminal
```

Configure host names as shown in the topology.

```
R1(config)# hostname R1
```

Configure interface IP addresses as shown in the IP Addressing Table.

```
R1(config)# interface g0/0/0
```

```
R1(config-if)# ip address 10.1.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
R1(config)# interface g0/0/1
```

```
R1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
interface GigabitEthernet3/0
ip address 10.1.1.1 255.255.255.252
negotiation auto
!
interface GigabitEthernet4/0
ip address 192.168.1.1 255.255.255.0
negotiation auto
!
interface GigabitEthernet5/0
no ip address
shutdown
negotiation auto
!
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.255 area 0
!
```

- b. To prevent the router from attempting to translate incorrectly entered commands as though they were host names, disable DNS lookup. R1 is shown here as an example.

```
R1(config)# no ip domain-lookup
no ip domain lookup
```

Step 3: Configure OSPF routing on the routers.

- a. Use the **router ospf** command in global configuration mode to enable OSPF on R1.

```
R1(config)# router ospf 1
```

- b. Configure the **network** statements for the networks on R1. Use an area ID of 0.

```
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

```
router ospf 1
passive-interface GigabitEthernet4/0
network 10.1.1.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.255 area 0
```

- c. Configure OSPF on R2 and R3.

```
R2(config)# router ospf 1
R2(config-router)# network 10.1.1.0 0.0.0.3 area 0
R2(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

```
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
```

```
R3(config)# router ospf 1
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

```
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
router ospf 1
  passive-interface GigabitEthernet4/0
  network 10.2.2.0 0.0.0.3 area 0
  network 192.168.3.0 0.0.0.255 area 0
```

- d. Issue the **passive-interface** command to change the G0/0/1 interface on R1 and R3 to passive.

```
R1(config)# router ospf 1
R1(config-router)# passive-interface g0/0/1

R1(config)#router ospf 1
R1(config-router)#pas
R1(config-router)#passive-interface g/4
^
% Invalid input detected at '^' marker.

R1(config-router)#passive-interface g4/0

R3(config)# router ospf 1
R3(config-router)# passive-interface g0/0/1

R3(config)#router ospf 1
R3(config-router)#pass
R3(config-router)#passive-interface g4/0
```

Step 4: Verify OSPF neighbors and routing information.

- a. Issue the **show ip ospf neighbor** command to verify that each router lists the other routers in the network as neighbors.

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.2.2.2	1	FULL/BDR	00:00:37	10.1.1.2	GigabitEthernet0/0/0

```
R1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address      Interface
10.2.2.2       1     FULL/BDR        00:00:39   10.1.1.2    GigabitEthernet3/0
```

- b. Issue the **show ip route** command to verify that all networks display in the routing table on all routers.

```
R1# show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
```

```
C       10.1.1.0/30 is directly connected, GigabitEthernet0/0/0
```

Lab - Securing the Router for Administrative Access

```
L      10.1.1.1/32 is directly connected, GigabitEthernet0/0/0
O      10.2.2.0/30 [110/2] via 10.1.1.2, 00:01:11, GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L      192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
•      192.168.3.0/24 [110/3] via 10.1.1.2, 00:01:07, GigabitEthernet0/0/0
```

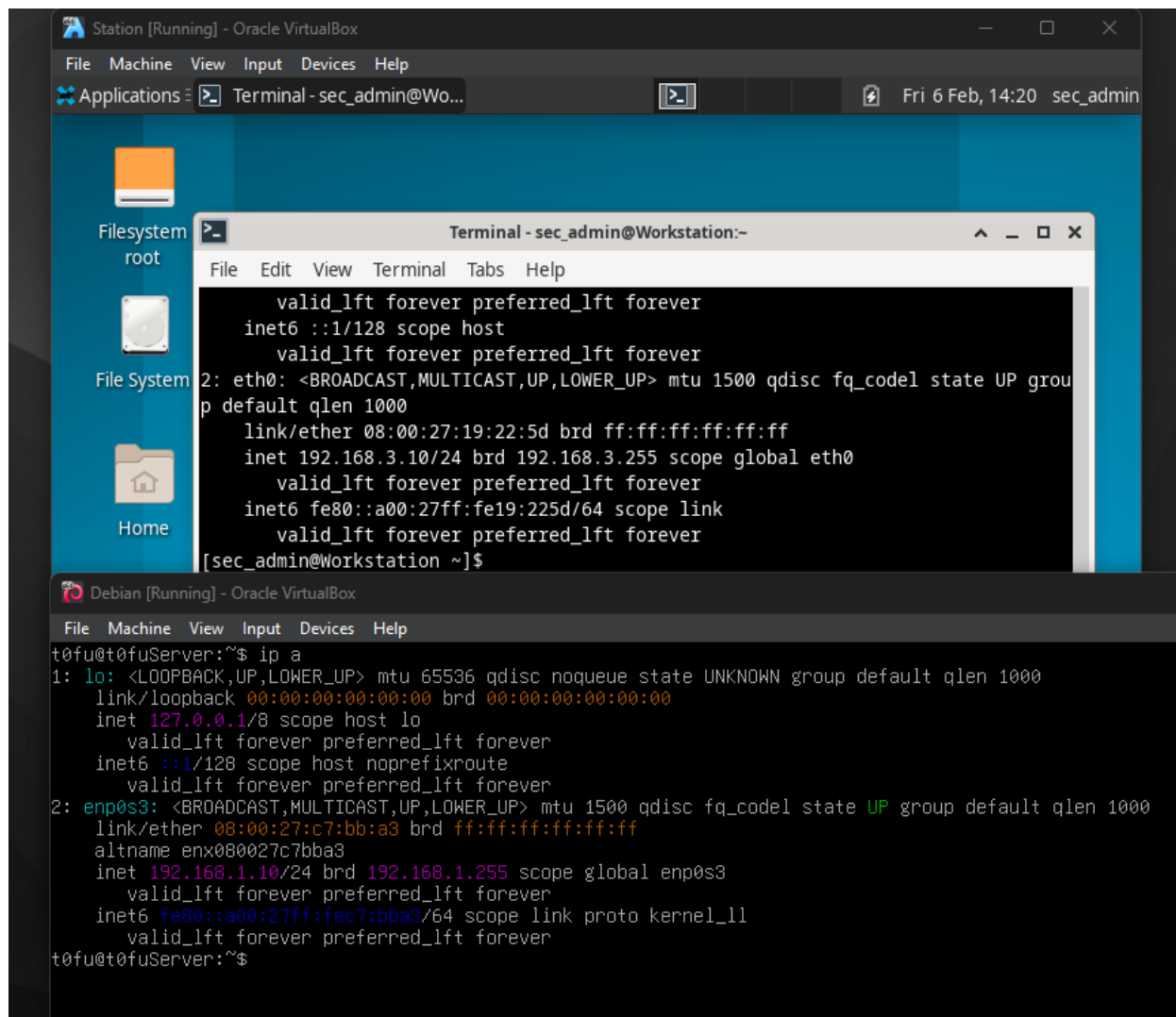
```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C      10.1.1.0/30 is directly connected, GigabitEthernet3/0
L      10.1.1.1/32 is directly connected, GigabitEthernet3/0
O      10.2.2.0/30 [110/2] via 10.1.1.2, 01:56:14, GigabitEthernet3/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, GigabitEthernet4/0
L      192.168.1.1/32 is directly connected, GigabitEthernet4/0
O      192.168.3.0/24 [110/3] via 10.1.1.2, 01:43:25, GigabitEthernet3/0
```

Step 5: Configure PC host IP settings.

Configure a static IP address, subnet mask, and default gateway for PC-A and PC-C as shown in the IP Addressing Table.



Step 6: Verify connectivity between PC-A and PC-C.

- a. Ping from R1 to R3.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

- b. Ping from PC-A, on the R1 LAN, to PC-C, on the R3 LAN.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

Note: If you can ping from PC-A to PC-C you have demonstrated that OSPF routing is configured and functioning correctly. If you cannot ping but the device interfaces are up and IP addresses are correct, use the **show run**, **show ip ospf neighbor**, and **show ip route** commands to help identify routing protocol-related problems.

```
Debian [Running] - Oracle VirtualBox
File Machine View Input Devices Help
t0fu@t0fuServer:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c7:bb:a3 brd ff:ff:ff:ff:ff:ff
    altname enx080027c7bba3
    inet 192.168.1.10/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fec7:bba3/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
t0fu@t0fuServer:~$ ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=255 time=9.02 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=255 time=5.05 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=255 time=9.74 ms
64 bytes from 10.1.1.1: icmp_seq=4 ttl=255 time=7.62 ms
64 bytes from 10.1.1.1: icmp_seq=5 ttl=255 time=4.03 ms
64 bytes from 10.1.1.1: icmp_seq=6 ttl=255 time=9.40 ms
^C
--- 10.1.1.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 500ms
rtt min/avg/max/mdev = 4.029/7.477/9.743/2.197 ms
t0fu@t0fuServer:~$ ping 192.168.3.1
PING 192.168.3.1 (192.168.3.1) 56(84) bytes of data.
64 bytes from 192.168.3.1: icmp_seq=1 ttl=253 time=46.8 ms
64 bytes from 192.168.3.1: icmp_seq=2 ttl=253 time=41.2 ms
64 bytes from 192.168.3.1: icmp_seq=3 ttl=253 time=38.5 ms
64 bytes from 192.168.3.1: icmp_seq=4 ttl=253 time=47.2 ms
^C
--- 192.168.3.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 300ms
rtt min/avg/max/mdev = 38.460/43.415/47.177/3.719 ms
t0fu@t0fuServer:~$ ping 192.168.3.10
PING 192.168.3.10 (192.168.3.10) 56(84) bytes of data.
64 bytes from 192.168.3.10: icmp_seq=1 ttl=61 time=45.9 ms
64 bytes from 192.168.3.10: icmp_seq=2 ttl=61 time=42.2 ms
64 bytes from 192.168.3.10: icmp_seq=3 ttl=61 time=61.0 ms
64 bytes from 192.168.3.10: icmp_seq=4 ttl=61 time=56.6 ms
64 bytes from 192.168.3.10: icmp_seq=5 ttl=61 time=56.1 ms
^C
--- 192.168.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 400ms
rtt min/avg/max/mdev = 42.237/52.389/61.038/7.094 ms
t0fu@t0fuServer:~$
```

Step 7: Save the basic running configuration for each router.

Save the basic running configuration for the routers as text files on your PC. These text files can be used to restore configurations later in the lab.

Part 2: Configure SNMPv3 Security using an ACL.

Simple Network Management Protocol (SNMP) enables network administrators to monitor network performance, manage network devices, and troubleshoot network problems. SNMPv3 provides secure access by authenticating and encrypting SNMP management packets over the network. You will configure SNMPv3 using an ACL on R1.

Step 1: Configure an ACL on R1 that will restrict access to SNMP on the 192.168.1.0 LAN.

- a. Create a standard access-list named **PERMIT-SNMP**.

```
R1(config)# ip access-list standard PERMIT-SNMP
```

- b. Add a permit statement to allow only packets on R1's LAN.

```
R1(config-std-nacl)# permit 192.168.1.0 0.0.0.255
```

```
R1(config-std-nacl)# exit
```

Step 2: Configure the SNMP view.

Configure a SNMP view called **SNMP-RO** to include the ISO MIB family.

```
R1(config)# snmp-server view SNMP-RO iso included
```

Step 3: Configure the SNMP group.

Call the group name **SNMP-G1**, and configure the group to use SNMPv3 and require both authentication and encryption by using the **priv** keyword. Associate the view you created in Step 2 to the group, giving it read only access with the **read** parameter. Finally specify the ACL **PERMIT-SNMP**, configured in Step 1, to restrict SNMP access to the local LAN.

```
R1(config)# snmp-server group SNMP-G1 v3 priv read SNMP-RO access PERMIT-SNMP
```

Step 4: Configure the SNMP user.

Configure an **SNMP-Admin** user and associate the user to the **SNMP-G1** group you configured in Step 3. Set the authentication method to **SHA** and the authentication password to **Authpass**. Use AES-128 for encryption with a password of **Encrypass**.

```
R1(config)# snmp-server user SNMP-Admin SNMP-G1 v3 auth sha Authpass priv aes 128 Encrypass
```

```
R1(config)# end
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip acc
R1(config)#ip acces
R1(config)#ip access-list stan
R1(config)#ip access-list standard PER
R1(config)#ip access-list standard PERMIT-SNMP
R1(config-std-nacl)#permit 192.168.1.0 0.0.0.255
R1(config-std-nacl)#exit
R1(config)#snmp-server view SNMP-RO iso included
R1(config)#snm
R1(config)#snmp-
R1(config)#snmp-server g
R1(config)#snmp-server group SNMP-G1 v3 priv read SNMP-RO access PERMIT-SNMP
R1(config)#$-Admin SNMP-G1 v3 auth sha Authpass priv aes 128 Encrypass
R1(config)#
*Feb  7 10:07:47.407: Configuring snmpv3 USM user, persisting snmpEngineBoots. Please Wait...

R1(config)#end
R1#
*Feb  7 10:07:52.679: %SYS-5-CONFIG_I: Configured from console by console
```

Step 5: Verify your SNMP configuration.

- Use the **show snmp group** command in privilege EXEC mode to view the SNMP group configuration. Verify that your group is configured correctly.

Note: If you need to make changes to the group, use the command **no snmp group** to remove the group from the configuration and then re-add it with the correct parameters.

```
R1# show snmp group
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: SNMP-G1                  security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
readview : SNMP-RO                  writeview: <no writeview specified>
notifyview: <no notifyview specified>
row status: active                  access-list: PERMIT-SNMP
```

```
R1#show snmp group
groupname: ILMI                      security model:v1
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                      security model:v2c
contextname: <no context specified>  storage-type: permanent
readview : *ilmi                     writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: SNMP-G1                  security model:v3 priv
contextname: <no context specified>  storage-type: nonvolatile
readview : SNMP-RO                  writeview: <no writeview specified>
notifyview: <no notifyview specified>
row status: active                  access-list: PERMIT-SNMP
```

- Use the command **show snmp user** to view the SNMP user information.

Note: The **snmp-server user** command is hidden from view in the configuration for security reasons. However, if you need to make changes to a SNMP user, you can issue the command **no snmp-server user** to remove the user from the configuration, and then re-add the user with the new parameters.

```
R1# show snmp user

User name: SNMP-Admin
Engine ID: 8000000903007079B3923640
```

```
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: SNMP-G1
```

```
R1#show snmp user

User name: SNMP-Admin
Engine ID: 800000090300CA0113B90000
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: SNMP-G1
```

Part 3: Configure a Synchronized Time Source Using NTP.

R2 will be the master NTP clock source for routers R1 and R3.

Note: R2 could also be the master clock source for switches S1 and S3, but it is not necessary to configure them for this lab.

Step 1: Set Up the NTP Master using Cisco IOS commands.

R2 is the master NTP server in this lab. All other routers and switches learn the time from it, either directly or indirectly. For this reason, you must ensure that R2 has the correct Coordinated Universal Time set.

- a. Use the **show clock** command to display the current time set on the router.

```
R2# show clock
*18:18:25.443 UTC Sun Jan 31 2021
```

- b. To set the time on the router, use the **clock set time** command.

```
R2# clock set 11:17:00 Jan 31 2021
R2#
*Jan 31 11:17:00.001: %SYS-6-CLOCKUPDATE: System clock has been updated from
18:19:03 UTC Sun Jan 31 2021 to 11:17:00 UTC Sun Jan 31 2021, configured from
console by console.
Jan 31 11:17:00.001: %PKI-6-AUTHORITATIVE_CLOCK: The system clock has been
set.
```

- c. Configure NTP authentication by defining the authentication key number, hashing type, and password that will be used for authentication. The password is case sensitive.

```
R2# config t
R2(config)# ntp authentication-key 1 md5 NTPpassword
```

- d. Configure the trusted key that will be used for authentication on R2.

```
R2(config)# ntp trusted-key 1
```

- e. Enable the NTP authentication feature on R2.

```
R2(config)# ntp authenticate
```

- f. Configure R2 as the NTP master using the **ntp master stratum-number** command in global configuration mode. The stratum number indicates the distance from the original source. For this lab, use a stratum number of **3** on R2. When a device learns the time from an NTP source, its stratum number becomes one greater than the stratum number of its source.

```
R2(config)# ntp master 3
```

```
R2#show clock
*10:13:03.239 UTC Sat Feb 7 2026
R2#clock set 11:17:00 Jan 31 2021
R2#
*Jan 31 11:17:00.003: %SYS-6-CLOCKUPDATE: System clock has been
021, configured from console by console.
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ntp authentication-key 1 md5 NTPpassword
R2(config)#ntp trusted-key 1
R2(config)#ntp authenticate
R2(config)#ntp authenticate
R2(config)#ntp master 3
```

Step 2: Configure R1 and R3 as NTP clients using the CLI.

- a. Issue the **debug ntp all** command to see NTP activity on R1 as it synchronizes with R2. Review the debug messages as you proceed through this step.

```
R1# debug ntp all
NTP events debugging is on
NTP core messages debugging is on
NTP clock adjustments debugging is on
NTP reference clocks debugging is on
NTP packets debugging is on
```

```
R2#debug ntp all
NTP events debugging is on
NTP core messages debugging is on
NTP clock adjustments debugging is on
NTP reference clocks debugging is on
NTP packets debugging is on
```

- b. Configure NTP authentication by defining the authentication key number, hashing type, and password that will be used for authentication.

```
R1# config t
R1(config)# ntp authentication-key 1 md5 NTPpassword
R1(config)#
*Jan 31 18:41:23.707: NTP Core(INFO): keys initilized.
*Jan 31 18:41:23.712: NTP Core(NOTICE): proto: precision = usec
*Jan 31 18:41:23.712: %NTP : Drift Read Failed (String Error).
*Jan 31 18:41:23.712: NTP Core(DEBUG): drift value read: 0.000000000
*Jan 31 18:41:23.712: NTP Core(NOTICE): ntpd PPM
*Jan 31 18:41:23.712: NTP Core(NOTICE): trans state : 1
*Jan 31 18:41:23.712: NTP: Initialized interface GigabitEthernet0/0/0
*Jan 31 18:41:23.712: NTP: Initialized interface GigabitEthernet0/0/1
*Jan 31 18:41:23.712: NTP: Initialized interface LIINO
R1(config)#
*Jan 31 18:41:23.713: NTP Core(INFO): more memory added for keys.
*Jan 31 18:41:23.713: NTP Core(INFO): key (1) added.
```

- c. Configure the trusted key that will be used for authentication. This command provides protection against accidentally synchronizing the device to a time source that is not trusted.

```
R1(config)# ntp trusted-key 1
```

```
R1(config)#  
*Jan 31 18:43:56.191: NTP Core(INFO): key (1) marked as trusted.
```

- d. Enable the NTP authentication feature.

```
R1(config)# ntp authenticate  
R1(config)#  
*Jan 31 18:44:33.482: NTP Core(INFO): 0.0.0.0 C01C 0C clock_step
```

- e. R1 and R3 will become NTP clients of R2. Use the command **ntp server *hostname***. The host name can also be an IP address.

Note: The command **ntp update-calendar** may be necessary to periodically updates the calendar with the NTP time for other IOS images.

```
R1(config)# ntp server 10.1.1.2  
R1(config)#  
*Jan 31 18:45:29.714: NTP message sent to 10.1.1.2, from interface  
'GigabitEthernet0/0/0' (10.2.2.1).  
*Jan 31 18:45:29.715: NTP message received from 10.1.1.2 on interface  
'GigabitEthernet0/0/0' (10.2.2.1).  
*Jan 31 18:45:29.716: NTP Core(DEBUG): ntp_receive: message received  
*Jan 31 18:45:29.716: NTP Core(DEBUG): ntp_receive: peer is 0x80007FA135BB32F8, next  
action is 1.  
*Jan 31 18:45:29.716: NTP Core(DEBUG): Peer becomes reachable, poll set to 6.  
  
*Jan 31 18:45:29.716: NTP Core(INFO): 10.1.1.2 8014 84 reachable  
*Jan 31 18:45:29.716: NTP Core(INFO): 10.1.1.2 962A 8A sys_peer  
R1(config)#  
*Jan 31 18:45:29.716: NTP: step(0xFFFF9D56.B5A1C9F4): local_offset =  
0x00000000.00000000, curtime = 0xE3C17949.B74BC8A0  
*Jan 31 11:44:32.426: NTP Core(NOTICE): time reset -25257.290500 s  
*Jan 31 11:44:32.426: NTP Core(NOTICE): trans state : 4  
*Jan 31 11:44:32.426: NTP Core(INFO): 0.0.0.0 C62C 0C clock_step  
*Jan 31 11:44:32.426: NTP Core(INFO): 0.0.0.0 C03C 0C clock_step  
*Jan 31 11:44:33.423: NTP message sent to 10.1.1.2, from interface  
'GigabitEthernet0/0/0' (10.2.2.1).  
*Jan 31 11:44:33.424: NTP message received from 10.1.1.2 on interface  
'GigabitEthernet0/0/0' (10.2.2.1).  
*Jan 31 11:44:33.424: NTP Core(DEBUG): ntp_receive: message received  
*Jan 31 11:44:33.424: NTP Core(DEBUG): ntp_receive: peer is 0x80007FA135BB32F8, next  
action is 1.  
*Jan 31 11:44:33.424: NTP Core(DEBUG): Peer becomes reachable, poll set to 6.  
  
*Jan 31 11:44:33.424: NTP Core(INFO): 10.1.1.2 8034 84 reachable  
*Jan 31 11:44:33.425: NTP Core(INFO): 10.1.1.2 964A 8A sys_peer
```

- f. Issue the **undebug all** or the **no debug ntp all** command to turn off debugging.

```
R1# undebug all
```

```

Jan 31 11:19:54.711: NTP message sent to 10.1.1.1, from interface 'GigabitEthernet3/0' (10.1.1.2).
R2#
Jan 31 11:19:56.707: NTP message received from 10.1.1.1 on interface 'GigabitEthernet3/0' (10.1.1.2).
Jan 31 11:19:56.707: NTP Core(DEBUG): ntp_receive: message received
Jan 31 11:19:56.707: NTP Core(DEBUG): ntp_receive: peer is 0x00000000, next action is 3.
Jan 31 11:19:56.707: NTP message sent to 10.1.1.1, from interface 'GigabitEthernet3/0' (10.1.1.2).
R2#
Jan 31 11:19:58.707: NTP message received from 10.1.1.1 on interface 'GigabitEthernet3/0' (10.1.1.2).
Jan 31 11:19:58.711: NTP Core(DEBUG): ntp_receive: message received
Jan 31 11:19:58.711: NTP Core(DEBUG): ntp_receive: peer is 0x00000000, next action is 3.
Jan 31 11:19:58.711: NTP message sent to 10.1.1.1, from interface 'GigabitEthernet3/0' (10.1.1.2).
R2#
Jan 31 11:20:00.703: NTP message received from 10.1.1.1 on interface 'GigabitEthernet3/0' (10.1.1.2).
Jan 31 11:20:00.703: NTP Core(DEBUG): ntp_receive: message received
Jan 31 11:20:00.703: NTP Core(DEBUG): ntp_receive: peer is 0x00000000, next action is 3.
Jan 31 11:20:00.703: NTP message sent to 10.1.1.1, from interface 'GigabitEthernet3/0' (10.1.1.2).
R2#
Jan 31 11:20:02.707: NTP message received from 10.1.1.1 on interface 'GigabitEthernet3/0' (10.1.1.2).
Jan 31 11:20:02.707: NTP Core(DEBUG): ntp_receive: message received
Jan 31 11:20:02.707: NTP Core(DEBUG): ntp_receive: peer is 0x00000000, next action is 3.
Jan 31 11:20:02.707: NTP message sent to 10.1.1.1, from interface 'GigabitEthernet3/0' (10.1.1.2).
R2#ubdeb
R2#ubde
R2#und
R2#undeg
R2#undeb
R2#undebg all
All possible debugging has been turned off

```

- g. Verify that R1 has made an association with R2 with the **show ntp associations** command. You can also use the more verbose version of the command by adding the **detail** argument. It might take some time for the NTP association to form.

R1# **show ntp associations**

```

address      ref clock      st  when  poll reach  delay  offset  disp
~10.1.1.2    127.127.1.1    3   14    64      3  0.000 -280073 3939.7
*sys.peer, # selected, +candidate, -outlyer, x falseticker, ~ configured

```

- h. Verify the time on R1 after it has made an association with R2.

R1# **show clock**

```
*11:49:27.709 UTC Sun Jan 31 2021
```

- i. Repeat the NTP configurations to configure R3 as an NTP client.

```

R1#show ntp associations
*Feb 7 10:17:04.143: %SYS-5-CONFIG_I: Configured from console by console
R1#show ntp associations

address      ref clock      st  when  poll reach  delay  offset  disp
~10.1.1.2    127.127.1.1    3   49    64      0  0.000  0.000 15937.
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R1#show clock
*10:17:17.027 UTC Sat Feb 7 2026

```

Part 4: Configure syslog Support on R1 and PC-A.

Step 1: Install and start the syslog server.

Free or trial versions of syslog server can be downloaded from the Internet. Use a web browser to search for “free windows syslog server” and refer to the software documentation for more information. Your instructor may also recommend a suitable syslog server for classroom use.

If a syslog server is not currently installed on the host, download a syslog server and install it on PC-A. If it is already installed, go to the next step.

Step 2: Configure R1 to log messages to the syslog server using the CLI.

- Start the syslog server.
- Verify that you have connectivity between R1 and PC-A by pinging the R1 G0/0/1 interface IP address 192.168.1.1. If it is not successful, troubleshoot as necessary before continuing.
- NTP was configured in a previous part to synchronize the time on the network. Displaying the correct time and date in syslog messages is vital when using syslog to monitor a network. If the correct time and date of a message is not known, it can be difficult to determine what network event caused the message.

Verify that the timestamp service for logging is enabled on the router using the **show run** command. Use the following command if the timestamp service is not enabled.

```
R1(config)# service timestamps log datetime msec
```

- Configure the syslog service on the router to send syslog messages to the syslog server.

```
R1(config)# logging host 192.168.1.3
```

Step 3: Configure the logging severity level on R1.

Logging traps can be set to support the logging function. A trap is a threshold that when reached, triggers a log message. The level of logging messages can be adjusted to allow the administrator to determine what kinds of messages are sent to the syslog server. Routers support different levels of logging. The eight levels range from 0 (emergencies), indicating that the system is unstable, to 7 (debugging), which sends messages that include router information.

Note: The default level for syslog is 6, informational logging. The default for console and monitor logging is 7, debugging.

- Use the **logging trap** command to determine the options for the command and the various trap levels available.

```
R1(config)# logging trap ?
<0-7>          Logging severity level
alerts         Immediate action needed          (severity=1)
critical       Critical conditions              (severity=2)
debugging      Debugging messages              (severity=7)
emergencies    System is unusable               (severity=0)
errors         Error conditions                 (severity=3)
informational  Informational messages           (severity=6)
notifications  Normal but significant conditions (severity=5)
warnings      Warning conditions                (severity=4)
<cr>
```

```
R1(config)#loggi
R1(config)#logging trap ?
<0-7>      Logging severity level
alerts      Immediate action needed      (severity=1)
critical     Critical conditions            (severity=2)
debugging    Debugging messages                    (severity=7)
emergencies  System is unusable                   (severity=0)
errors       Error conditions                     (severity=3)
informational Informational messages                (severity=6)
notifications Normal but significant conditions    (severity=5)
warnings     Warning conditions                   (severity=4)
<cr>
```

- b. Define the level of severity for messages sent to the syslog server. To configure the severity levels, use either the keyword or the severity level number (0–7).

Severity Level	Keyword	Meaning
0	emergencies	System is unusable
1	alerts	Immediate action required
2	critical	Critical conditions
3	errors	Error conditions
4	warnings	Warning conditions
5	notifications	Normal but significant condition
6	informational	Informational messages
7	debugging	Debugging messages

Note: The severity level includes the level specified and anything with a lower severity number. For example, if you set the level to 4, or use the keyword **warnings**, you capture messages with severity level 4, 3, 2, 1, and 0.

- c. Use the **logging trap** command to set the severity level for R1.

```
R1(config)# logging trap warnings
```

What is the problem with setting the level of severity too high or too low?

Setting it too high (lowest level number) could generate logs that missed some very useful but not critical messages. Setting it too low (highest level number) could generate a large number of messages and fill up the logs with unnecessary information

If the command **logging trap critical** were issued, which severity levels of messages would be logged?

Emergencies, alerts, and critical messages

Step 4: Display the current status of logging for R1.

- a. Use the **show logging** command to see the type and level of logging enabled.

```
R1# show logging
```

```
Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0
overruns, xml disabled, filtering disabled)
```

```
No Active Message Discriminator.
```

```
No Inactive Message Discriminator.
```



```
Console logging: level debugging, 72 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 0 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 72 messages logged, xml disabled,
                  filtering disabled
Exception Logging: size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled
```

No active filter modules.

```
Trap logging: level warnings, 54 message lines logged
Logging to 192.168.1.3  (udp port 514, audit disabled,
                        link up),
                        3 message lines logged,
                        0 message lines rate-limited,
                        0 message lines dropped-by-MD,
                        xml disabled, sequence number disabled
                        filtering disabled
```

Logging Source-Interface: VRF Name:

<output omitted>

```
R1#show logging
Syslog logging: enabled (0 messages dropped, 16 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)
No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 52 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 0 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 66 messages logged, xml disabled,
                  filtering disabled
Exception Logging: size (8192 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

Trap logging: level warnings, 67 message lines logged
Logging Source-Interface:            VRF Name:

Log Buffer (8192 bytes):
*Feb  7 10:05:14.079: %IFMGR-7-NO_IFINDEX_FILE: Unable to open nvram:/ifIndex-table No such file or directory
```

At what level is console logging enabled?

Level 7 – debugging

At what level is trap logging enabled?

Level 4 – warnings

What is the IP address of the syslog server?

192.168.1.11

What port is syslog using?

udp port 514

Step 5: Make changes to the router and monitor syslog results on the PC.

- Verify that the syslog server is already started on PC-A. Start the server as necessary.
- To verify that syslog server is logging the message, disable and enable R1's G0/0/0 interface.

```
R1(config)# interface g0/0/0
R1(config-if)# shut
.Jan 31 12:02:50.376: %LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to administratively down
.Jan 31 12:02:51.376: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down
R1(config-if)# no shut
.Jan 31 12:03:11.302: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.4 port 514 started - CLI initiated
.Jan 31 12:03:14.365: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
.Jan 31 12:03:15.365: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
.Jan 31 12:03:59.894: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done
```

```
R1(config)#int g3/0
R1(config-if)#shut
R1(config-if)#
*Feb  7 10:23:09.311: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on GigabitEthernet3/0 from FULL to DOWN, Neighbor Down: Interface down or detached
R1(config-if)#
*Feb  7 10:23:11.283: %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to administratively down
*Feb  7 10:23:12.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to down
R1(config-if)#no shut
R1(config-if)#
*Feb  7 10:23:31.095: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Feb  7 10:23:32.095: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
R1(config-if)#ebd
^
% Invalid input detected at '^' marker.

R1(config-if)#end
R1#
*Feb  7 10:23:36.091: %SYS-5-CONFIG_I: Configured from console by console
*Feb  7 10:23:36.819: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on GigabitEthernet3/0 from LOADING to FULL, Loading Done
```

- Navigate to PC-A to view the syslog messages.

Router Interface Summary Table

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)

Lab - Securing the Router for Administrative Access

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
4221	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
4300	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.